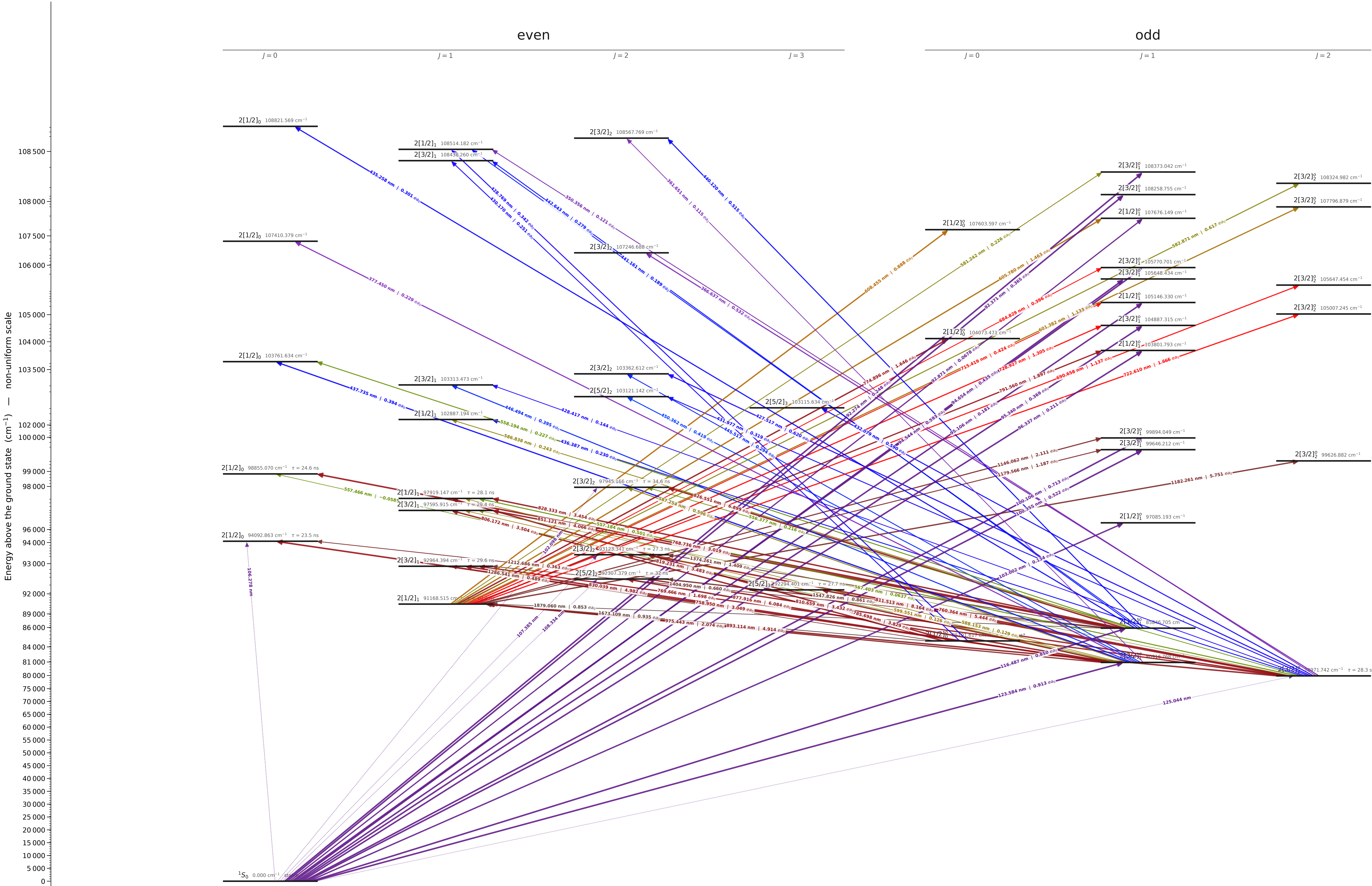
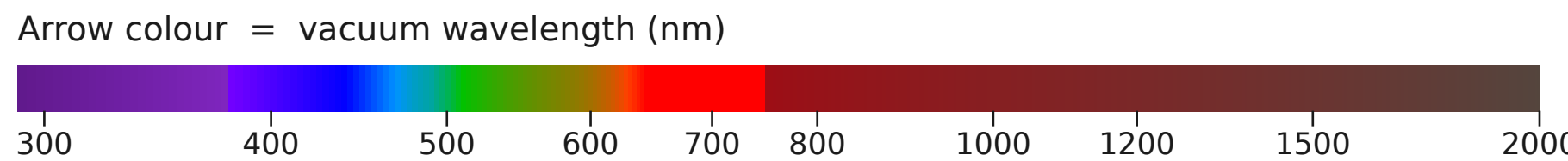


Krypton

Kr (Kr I). Strongest classified transitions below 2 μm (NIST): 45 levels, 84 lines.

Kr⁺ ionisation limit 112914.43 cm⁻¹ = 13.99961 eV (88.563 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle J || e r || J' \rangle$ in ea0
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\langle J e r J' \rangle$ (ea0)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
¹ S ₀ - 2[1/2] ^o	96.3374	-	3111.8995	0.211	3.350e+07	-	-	96.3 nm VUV line
¹ S ₀ - 2[3/2] ^o	100.1061	-	2994.7482	0.713	3.420e+08	-	-	100.1 nm VUV line
¹ S ₀ - 2[3/2] ₂	102.0979	-	2936.3222	-	-	4.6	-	TALIF calibration line: two-photon excitation at 2 x 204...
¹ S ₀ - 2[1/2] ₀	106.2780	-	2820.8331	-	-	6.773	-	two-photon line
¹ S ₀ - 2[3/2] ₂	107.3845	-	2791.7675	-	-	5.82	-	two-photon excitation
¹ S ₀ - 2[5/2] ₂	108.3337	-	2767.3056	-	-	4.967	-	two-photon line
¹ S ₀ - 2[3/2] ^o	123.5838	-	2425.8237	0.913	2.980e+08	-	-	123.6 nm resonance line: first
¹ S ₀ - 2[3/2] ^o	125.0442	-	2397.4925	-	3.540e-02	5.624e-09	100	125.0 nm M2 decay of the metastable 3P2 state
2[3/2] ^o - 2[3/2] ₂	587.2543	587.0916	510.4985	0.596	7.100e+05	4.6	2	587.1 nm fluorescence line detected in krypton TALIF
2[1/2] ₁ - 2[1/2] ^o	605.7803	605.6126	494.8864	1.463	6.500e+06	-	-	605.78 nm line of 86Kr
2[3/2] ^o - 2[3/2] ₂	760.3638	760.1545	394.2750	5.444	2.732e+07	5.82	75.1	760.2 nm: decay channel that feeds the metastable 5s[3/2]...
2[3/2] ^o - 2[5/2] ₃	811.5131	811.2900	369.4240	8.164	3.610e+07	5.747	100	811.3 nm
2[3/2] ^o - 2[3/2] ₂	819.2308	819.0056	365.9438	3.483	8.940e+06	5.82	24.1	819 nm second step of the optical
2[1/2] ^o - 2[3/2] ₂	826.5514	826.3242	362.7028	6.899	3.416e+07	4.6	97.7	826.3 nm: main decay branch

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
¹ S ₀ - 2[1/2] ^o	96.337	84-86	-206(15)
¹ S ₀ - 2[1/2] ^o	96.337	84-82	226(10)
¹ S ₀ - 2[1/2] ^o	96.337	84-80	475(15)
¹ S ₀ - 2[1/2] ^o	96.337	84-78	709(15)
¹ S ₀ - 2[3/2] ^o	100.106	84-86	-112(15)
¹ S ₀ - 2[3/2] ^o	100.106	84-83	91(14)
¹ S ₀ - 2[3/2] ^o	100.106	84-82	157(10)
¹ S ₀ - 2[3/2] ^o	100.106	84-80	284(15)
¹ S ₀ - 2[3/2] ^o	100.106	84-78	425(15)
¹ S ₀ - 2[1/2] ₀	106.278	84-86	-138(15)
¹ S ₀ - 2[1/2] ₀	106.278	84-83	88(17)
¹ S ₀ - 2[1/2] ₀	106.278	84-82	138(17)
¹ S ₀ - 2[1/2] ₀	106.278	84-80	311(16)
¹ S ₀ - 2[1/2] ₀	106.278	84-78	426(15)
¹ S ₀ - 2[3/2] ₂	107.385	84-86	-130(15)
¹ S ₀ - 2[3/2] ₂	107.385	84-83	83(16)
¹ S ₀ - 2[3/2] ₂	107.385	84-82	148(13)
¹ S ₀ - 2[3/2] ₂	107.385	84-80	295(14)
¹ S ₀ - 2[3/2] ₂	107.385	84-78	435(17)
¹ S ₀ - 2[5/2] ₂	108.334	84-86	-123(15)
¹ S ₀ - 2[5/2] ₂	108.334	84-83	95(15)
¹ S ₀ - 2[5/2] ₂	108.334	84-82	150(13)

Shift = ν(first isotope) - ν(second isotope).

⁸³Kr hyperfine structure (I = 9/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
2[3/2] ^o	-244.04(3)	-453.4(6)	F=5/2: +2476.65 F=7/2: +1820.86 F=9/2: +864.35 F=11/2: -477.87 F=13/2: -2309.70
2[3/2] ^o	-160.500	-105.900	F=7/2: +834.21 F=9/2: +231.10 F=11/2: -748.73
2[1/2] ₁	-143.00(9)	-	F=7/2: +786.50 F=9/2: +143.00 F=11/2: -643.50
2[5/2] ₃	-103.910(20)	-437.0(7)	F=3/2: +1514.21 F=5/2: +1345.48 F=7/2: +1083.77 F=9/2: +703.58 F=11/2: +172.14 F=13/2: -550.62 F=15/2: -1512.04
2[5/2] ₂	-156.49(8)	-407.7(13)	F=5/2: +1534.53 F=7/2: +1165.18 F=9/2: +588.38 F=11/2: -272.31 F=13/2: -1510.34
2[3/2] ₂	-108.400	-86.000	F=5/2: +1152.98 F=7/2: +811.21 F=9/2: +350.28 F=11/2: -245.92 F=13/2: -997.10

Measured A, B constants; sources in the data file.